

Can Deficits Finance Themselves?

Angeletos, Lian & Wolf (2023)

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Roadmap

- 1 Introduction
- 2 Model
- 3 Equilibrium
- 4 The Self-Financing Result
- 5 Quantitative Analysis
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Introduction

- Deficit goes up today, how can Debt/GDP be stable in the long-run?
- Conventional answer: implement fiscal adjustments eventually.
- Recently literature focused on channels that don't involve rising taxes.
- This paper will focus on a **self-financing channel**. The idea is that deficits could trigger a Keynesian boom, and that could raise revenue (**tax base channel**) and generate a surge in inflation which would erode the initial nominal government debt (**debt erosion channel**).

- The ingredients in our model will be: 1) a violation of the Ricardian Equivalence, and 2) the presence of nominal rigidities. Then, deficits will generate a Keynesian boom. The more the fiscal adjustment is postponed, the more persistent the boom is.
- **How does self-financing vary with the timing of tax hikes? Is complete self-financing even possible? Are these channels empirically relevant?**
- The take of the paper: This self-financing is more potent the more the fiscal adjustment is pushed into the future. Complete self-financing is theoretically possible. The mechanism is quantitatively relevant. From the two, the tax base channel is more important.

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Overview of the Model

- Closed Economy.
- OLG model with probability of survival ω in every period and hence heterogeneity on the time horizons of agents (Blanchard, 1985). No altruism. We will nest the standard PIH framework when $\omega = 1$.
- The Aggregate Supply will match the textbook New Keynesian Model.
- Deficit shock at date 0 complemented with a fiscal rule that will compensate for it. The graduality of that adjustment mechanism will be a relevant variable in the model.
- Monetary policy is going to be neutral, in the sense that will keep the ex-ante real interest rate on the steady state. That is, it will not counteract the boom neither strengthen it.

Model - Consumers

- As mentioned before, each agent will face each period probability of survival of ω . No altruism between generations.
- Consumers will be able to save in a risk-free annuity provided by the government that will pay in expectation I_t every period. $\frac{I_t}{\omega}$ if we condition on survival.
- To have the same steady-state as in a RANK model we will assume:
 - Share of dividends will be the same for all agents.
 - A union will negotiate for all the agents yielding for them same working hours and salary (and hence income and taxes). [» Details](#)
 - There will exist a social system fund (perfectly balanced) that will generate that **in steady-state** asset holdings will be the same among agents.

- The objective function of agents will be:

$$\mathbb{E}_t \left[\sum_{k=0}^{\infty} (\beta\omega)^k [u(C_{i,t+k}) - v(L_{i,t+k})] \right]$$

With the standard functions: $u(C) = \frac{C^{1-\frac{1}{\sigma}}-1}{1-\frac{1}{\sigma}}$ and $v(L) = \chi \frac{L^{1+\frac{1}{\phi}}}{1+\frac{1}{\phi}}$

- Budget Constraint for agents will be:

$$A_{i,t+1} = \underbrace{\frac{l_t}{\omega}}_{\text{annuity}} \left(A_{i,t} + P_t \cdot \underbrace{(W_t L_{i,t} + Q_{i,t})}_{Y_{i,t}} - C_{i,t} - T_{i,t} + S_{i,t} \right)$$

- $S_{i,t}$ will be transfers between generations implied by the self-sustained social fund. We will set them to be $S^{\text{new}} = D^{\text{ss}}$ for newborns and $S^{\text{old}} = -\frac{1-\omega}{\omega} D^{\text{ss}}$ for old generations. [► Details](#)

Model - Consumption Function

- The log-linearized version of consumption function:

$$c_{i,t} = (1 - \beta\omega)(\tilde{a}_{i,t} + \mathbb{E}_t[\sum_{k=0}^{\infty} (\beta\omega)^k (y_{t+k} - t_{t+k})]) - \gamma \mathbb{E}_t[\sum_{k=0}^{\infty} (\beta\omega)^k r_{t+k}]$$

- By aggregating across households:

$$c_t = (1 - \beta\omega)(\tilde{a}_t + \mathbb{E}_t[\sum_{k=0}^{\infty} (\beta\omega)^k (y_{t+k} - t_{t+k})]) - \gamma \mathbb{E}_t[\sum_{k=0}^{\infty} (\beta\omega)^k r_{t+k}]$$

- Production side is the same as a textbook New Keynesian model.
- A continuum of monopolistically competitive retailers who set prices subject a standard Calvo friction, hire labor on spot market, and pay-out all their profits as dividends back to household.
- These plus our assumptions on household labor supply will boil down to a standard New Keynesian Philips Curve (NKPC).

Model - Government

- Government will have no public expenditure, only taxes and debt.
- The government budget constraint in nominal terms is:

$$\frac{1}{I_t} B_{t+1} = B_t - P_t T_t$$

- The government budget constraint in real terms is:

$$D_{t+1} = R_t(D_t - T_t) \left(\frac{\mathbb{E}_t[\Pi_{t+1}]}{\Pi_{t+1}} \right)$$

- Log-linearizing around the steady-state:

$$d_{t+1} = \underbrace{\frac{1}{\beta}(d_t - t_t) + \frac{D^{ss}}{\gamma^{ss}} r_t}_{\text{expected debt burden tomorrow}} + \underbrace{\frac{D^{ss}}{\gamma^{ss}} (\mathbb{E}_t[\pi_{t+1}] - \pi_{t+1})}_{\text{debt erosion due to inflation surprise}}$$

Model- Government: Two Fiscal Rules

- We will propose two fiscal rules.
- The first rule will use a scalar τ_d that will parametrize the speed of fiscal adjustment. Arbitrarily fast for $\tau_d \rightarrow 1^-$ and arbitrarily slow for $\tau_d \rightarrow 0^+$:

$$T_{i,t} = \tau_y Y_{i,t} + \bar{T} - \mathcal{E}_t + \tau_d (D_t - D^{SS} + \mathcal{E}_t)$$

- Here $\mathcal{E}_t$ is a mean-zero i.i.d. deficit shock. Log-linearizing we get:

$$t_t = \underbrace{\tau_y y_t}_{\text{tax base effect}} + \underbrace{\tau_d \cdot (d_t + \varepsilon_t)}_{\text{fiscal adjustment}} - \underbrace{\varepsilon_t}_{\text{deficit shock}}$$

Model-Government: Two Fiscal Rules

- The alternative fiscal rule will be a time-dependent version of the previous one. After log-linearization it gives us:

$$t_t = \begin{cases} \tau_y y_t - \varepsilon_0 & \text{if } t = 0 \\ \tau_y y_t & \text{if } t \in \{1, \dots, H-1\} \\ d_t & t \geq H \end{cases}$$

- Note that here we shut down any deficit shocks at dates $t \geq 1$ allowing us to focus cleanly on impulse-responses to a date 0 shock.
- In equilibrium a Lemma will establish interchangeability between the rules by asserting there will be a mapping between τ_d and H , where if $\tau_d \rightarrow 0$ then $H \rightarrow \infty$ and viceversa.

Model - Government: Monetary Policy

- Monetary authority will set the rate of interest on nominal bonds as:

$$l_t = R^{SS} \mathbb{E}_t \left[\frac{\Pi_{t+1}}{\Pi^{SS}} \right] \left(\frac{Y_t}{Y^{SS}} \right)^\phi$$

- After log-lin:

$$r_t = \phi y_t$$

- But we will make monetary policy be neutral in the sense that $\phi = 0$.

$$r_t = 0$$

- By neutral, we mean monetary policy neither offsets the fiscal stimulus nor does it increase fiscal space by allowing government's real borrowing cost to fall.

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Equilibrium - Aggregate Demand

- By using the previous log-linearized consumption function and including $r_t = 0$ plus market clearing conditions ($c_t = y_t$ and $\tilde{a}_t = d_t$) and the fiscal rule, we get a key equation of the paper. The aggregate demand equation:

$$y_t = \mathcal{F}_1 \cdot (d_t + \varepsilon_t) + \mathcal{F}_2 \cdot \mathbb{E}_t \left[(1 - \beta\omega) \sum_{k=0}^{\infty} (\beta\omega)^k (y_{t+k}) \right]$$

- Where:

$$\mathcal{F}_1 = \frac{(1 - \beta\omega)(1 - \omega)(1 - \tau_d)}{1 - \omega(1 - \tau_d)}$$

$$\mathcal{F}_2 = 1 - \frac{(1 - \omega)\tau_y}{1 - \omega(1 - \tau_d)}$$

- What happens if $\omega = 1$? What happens when $\omega < 1$ and $\tau_d < 1$?

Equilibrium- Aggregate Supply

- Supply side of the model is exactly its RANK counterpart.
- Labor supply is given by [Details](#):

$$\frac{1}{\varphi} \ell_t = w_t - \frac{1}{\sigma} c_t$$

- Following standard steps we can boil this down to the familiar NKPC:

$$\pi_t = \kappa y_t + \beta \mathbb{E}_t[\pi_{t+1}]$$

- Here $\kappa \geq 0$ depends on ξ (procyclicality of real marginal costs) and θ (Calvo reset probability).

Equilibrium-Government: Law of Motion for Public Debt

- By combining the government's flow budget constraint with the first fiscal rule we get to the following law of motion for public debt:

$$d_{t+1} = \beta^{-1} (d_t + \varepsilon_t - \tau_d \cdot (d_t + \varepsilon_t) - \tau_y y_t) - \frac{D^{SS}}{Y^{SS}} (\pi_{t+1} - \mathbb{E}_t[\pi_t])$$

- With the initial condition:

$$d_0 = -\frac{D^{SS}}{Y^{SS}} \pi_0$$

- Plus the usual no-Ponzi restriction $\lim_{k \rightarrow \infty} \mathbb{E}_t[\beta^k d_{t+k}] = 0$

Equilibrium Definition

- **Definition:** An equilibrium is a stochastic path $\{y_t, \pi_t, d_t\}_{t=0}^{\infty}$ for output, inflation and real value of public debt that is bounded and satisfies:

$$y_t = \mathcal{F}_1 \cdot (d_t + \varepsilon_t) + \mathcal{F}_2 \cdot \mathbb{E}_t \left[(1 - \beta\omega) \sum_{k=0}^{\infty} (\beta\omega)^k (y_{t+k}) \right]$$

$$\pi_t = \kappa y_t + \beta \mathbb{E}_t[\pi_{t+1}]$$

$$d_{t+1} = \beta^{-1} (d_t + \varepsilon_t - \tau_d \cdot (d_t + \varepsilon_t) - \tau_y y_t) - \frac{D^{SS}}{Y^{SS}} (\pi_{t+1} - \mathbb{E}_t[\pi_t])$$

- Plus the initial condition for real public debt and the No Ponzi Condition for the Government.

Equilibrium - Proposition 1)

- **Proposition 1:** Suppose that $\omega < 1$ and $\tau_y > 0$. There exists a unique (bounded) equilibrium. Along this equilibrium, real output and real debt satisfy:

$$y_t = \chi \cdot (d_t + \varepsilon_t) \text{ and } \mathbb{E}_t[d_{t+1}] = \rho_d \cdot (d_t + \varepsilon_t)$$

for some $\chi > 0$ and $\rho_d \in (0, 1)$. These coefficients solve the following fixed point problem:

$$\chi = \frac{\mathcal{F}_1}{1 - \mathcal{F}_2 \frac{1 - \beta\omega}{1 - \beta\omega\rho_d}}$$

$$\rho_d = \beta^{-1}(1 - \tau_d - \tau_y\chi)$$

Finally inflation will satisfy:

$$\pi_t = \frac{\kappa}{1 - \beta\rho_d} y_t$$

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The Self-Financing Results

- Iterating forward the law of motion of public debt and using the No Ponzi Condition we get:

$$\varepsilon_0 = \underbrace{\tau_d(\varepsilon_0 + \sum_{k=0}^{\infty} \beta^k \mathbb{E}_0[d_k])}_{\text{fiscal adjustment}} + \underbrace{\sum_{k=0}^{\infty} \tau_y \beta^k \mathbb{E}_0[y_k]}_{\text{self-fin. via tax base}} + \underbrace{\frac{D^{SS}}{Y^{SS}}(\pi_0 - \mathbb{E}_{-1}[\pi_0])}_{\text{self-fin. via debt-erosion}}$$

- We can thus define the degree of self-financing ν :

$$\nu = \frac{\sum_{k=0}^{\infty} \tau_y \beta^k \mathbb{E}_0[y_k] + \frac{D^{SS}}{Y^{SS}} \pi_0}{\varepsilon_0}$$

- Where we can decompose for the sources of self-financing:

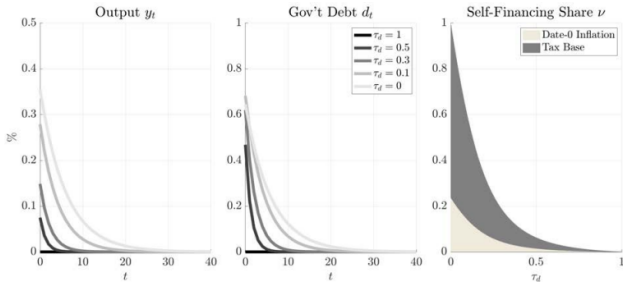
$$\nu = \nu_y + \nu_p.$$

► Other Results

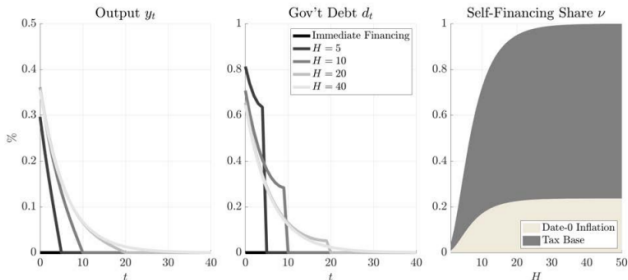
The Self-Financing Result

- **Theorem 1:** Suppose that $\omega < 1$ and $\tau_y > 0$, and let fiscal policy follow either our baseline rule or the variant rule. The equilibrium degree of self-financing, ν , has the following properties:
 - ① **Monotonicity:** It is increasing in the delay of fiscal adjustment; i.e., ν is decreasing in τ_d for the fiscal rule (6) and increasing in H for the fiscal rule (7).
 - ② **Limit:** It converges to 1 as fiscal adjustment is delayed further and further; i.e., $\nu \rightarrow 1$ (from below) and $\tau_d \rightarrow 0$ (from above), or as $H \rightarrow \infty$. Furthermore, these two limits induce the same equilibrium paths $\{y_t, \pi_t, d_t\}_{t=0}^{\infty}$ and in this common limit self-financing is sufficiently strong to return real government debt to steady state.

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Economic Intuition: Keynesian Cross

- Fully rigid prices model, with myopic consumer demand provided by:

$$c = MPC \cdot y_{disp}$$

- Output at 0 is fully demand-determined. By generating the fiscal deficit experiment at period 0 we get that output at period 0 is:

$$y = \frac{MPC}{1 - (1 - \tau_y)MPC} \cdot \varepsilon$$

- Then the degree of self-financing on this economy is:

$$\nu = \frac{\tau_y y}{\varepsilon} = \frac{\tau_y MPC}{1 - (1 - \tau_y)MPC}$$

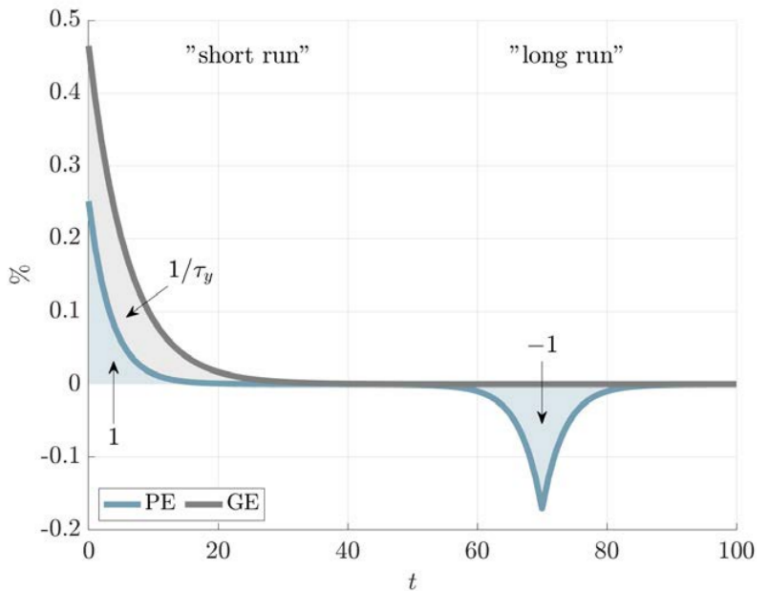
- Takeaways: 1) Larger MPC maps to larger self-financing.
2) in the limit if $MPC \rightarrow 1$, then $\nu \rightarrow 1$.

Economic Intuition: Back to the Dynamic Model

- To what extent was that simple model informative about our full dynamic economy?
- The result of $\nu \rightarrow 1$ relied on two properties: 1) expected date 1 tax hike does not affect date 0 spending, 2) the date 0 transfer and the additional income it creates are fully spent at date 0 ($MPC \rightarrow 1$).
- **Core intuition of the model:** as financial delay (H) increases the dynamic economy starts to emulate those two features of the static example.

Economic Intuition: Back to the Dynamic Model

- Let's define $\mathcal{M}_{t,t+\ell}$ as the $(t, t + \ell)$ element of the matrix of intertemporal MPCs.
- Two properties of our economy's aggregate consumption function are crucial here:
 - ① **Discounting:** $\beta^{-\ell} \mathcal{M}_{t,t+\ell}$ is strictly decreasing in ℓ , and in the limit it goes to 0 as ℓ goes to ∞ . In plain words, an income change of a fixed present value that occurs further and further into the future has a diminishing and eventually vanishing effect on consumption. This is **not** true in the PIH benchmark.
 - ② **Front-loading:** $\mathcal{M}_{t+\ell,t}$ is decreasing in ℓ and in the limit goes to 0 when ℓ goes to ∞ . In plain words, any income shock today has a vanishing effect on consumption far enough in the future. Consumers will spend their income faster than in the PIH benchmark.



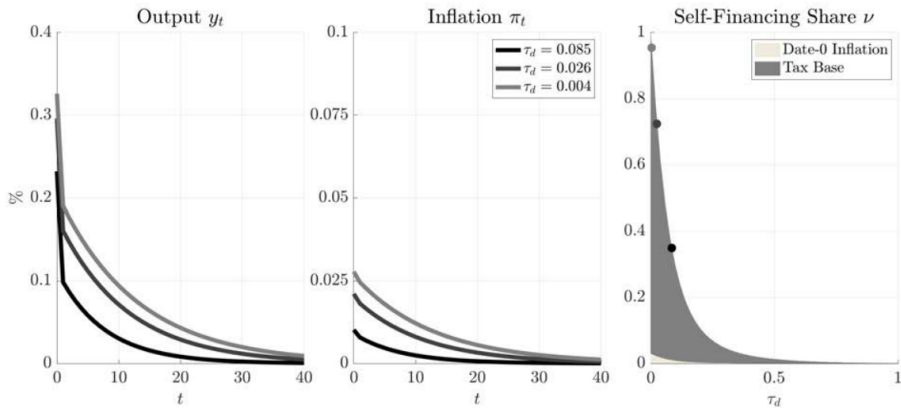
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Calibration

Parameter	Description	Value	Target	Value
<i>Consumer spending</i>				
μ	Share of HtM	0.111	Average MPC	0.22
ω	OLG survival rate	0.88	Short-run MPC slope	0.88
<i>Fiscal adjustment</i>				
τ_d	Tax feedback	{0.085,0.026,0.004}	Literature, see text	
<i>Rest of the model</i>				
σ	EIS	1	Standard	
β	Discount factor	0.998	Annual real rate	0.01
τ_y	Tax rate	0.3	Average Labor Tax	0.3
D^{ss}/Y^{ss}	Gov't debt	1.04	Liq. wealth holdings	1.04
κ	NKPC slope	0.01	Standard	

Table 1: Hybrid OLG-spender model, calibration.



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Conclusions

- In environments where the Ricardian Equivalence does not hold and there exist nominal rigidities when fiscal authorities delay the tax adjustments, demand-led boom gets larger and more persistent contributing a greater share of self-financing.
- In the limit, deficits can even fully pay for themselves.
- Practically, a meaningful degree of self-financing is empirically plausible.
- In the quantitative exercise ν showed a large range between 0.3 and 0.9. We conclude this channel can be potentially very large.
- The tax base channel is the most relevant of the two self-financing channels.

Unions and Labor/Income Homogeneity

- We want to abstract from the heterogeneity of labor supply.
- Assume that labor supply is intermediated by labor unions. Those unions bargain on behalf of households, equalizing the (post-tax) real wage and the MRS between consumption and labor supply (for the aggregate consumption).
- It will follow that all households work the same hours and receive same labor income.

$$(1 - \tau_y) W_t = \frac{\chi \int_0^1 L_t^{\frac{1}{\varphi}} di}{\int_0^1 C_{i,t}^{-1/\sigma} di}$$

- By log-lin:

$$\frac{1}{\varphi} \ell_t = w_t - \frac{1}{\sigma} c_t$$

Algebra of the Social Fund

- To get that steady state asset holdings are equal across generations we need that $\tilde{A}_{new}^{SS} = \tilde{A}_{old}^{SS}$.
- By using previous formulas, plus the fact the social fund has to be perfectly balanced:

$$\tilde{A}_{new}^{SS} = \frac{I^{SS}}{\omega} \cdot (-T^{SS} + S_{new})$$

$$\tilde{A}_{old}^{SS} = \frac{\omega}{\omega - I^{SS}} \frac{I^{SS}}{\omega} \cdot \left(-T^{SS} - \frac{1 - \omega}{\omega} S_{new} \right)$$

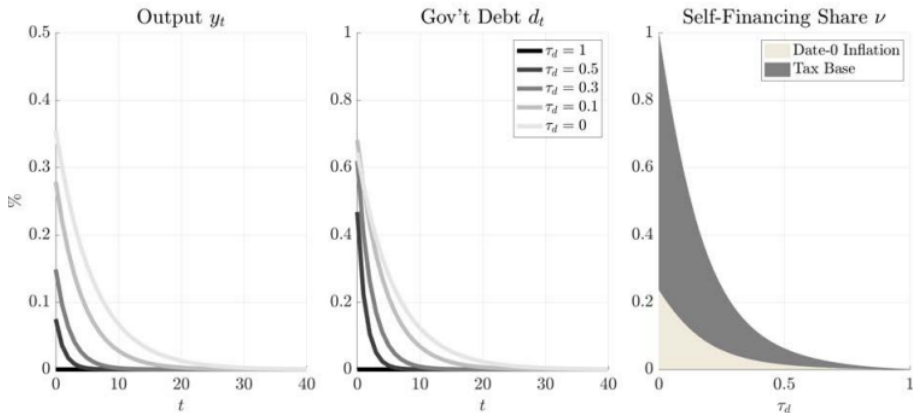
- By equating both and using that $T^{SS} = \frac{I^{SS}-1}{I^{SS}} D^{SS}$ (from the government flow budget constraint) we get the desired result:

$$S_{new} = D^{SS} \text{ and } S_{old} = -\frac{1 - \omega}{\omega} D^{SS}$$

Other Self-Financing Results

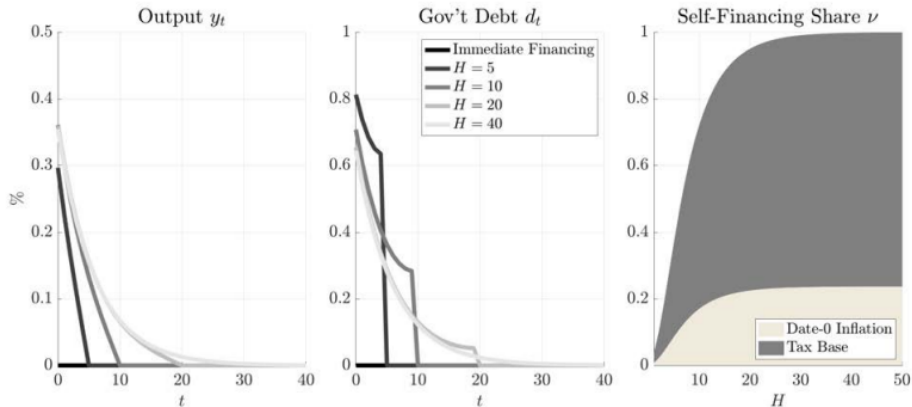
- **Proposition 2:** Let $\omega < 1$, $\tau_y > 0$, $\kappa > 0$ and $\frac{D^{SS}}{Y^{SS}} > 0$. Then ν is increasing in κ and in $\frac{D^{SS}}{Y^{SS}}$. Also $\frac{\nu_p}{\nu}$ is increasing in κ and in $\frac{D^{SS}}{Y^{SS}}$.
- **Proposition 3:** Let $\omega < 1$, $\tau_y > 0$, $\kappa = 0$. Then, ν is increasing in τ_y . With $\kappa > 0$, this might not hold if κ and/or $\frac{D^{SS}}{Y^{SS}}$ are large enough. That will not be the case for the parametrization of the quantitative exercise.
- **Proposition 4:** Let $\omega < 1$ and $\tau_y > 0$. Focus on the first fiscal rule.
 - For any $\tau_d > 0$, lower ω raises ν and decreases ρ_d . So larger departure from PIH yields both larger and faster self-financing.
 - Let ρ_d^{full} be the limit of ρ_d as $\tau_d \rightarrow 0$. Then, for any $\omega < 1$, $\rho_d^{full} < 1$. Also $\rho_d^{full} \rightarrow 1$ as $\omega \rightarrow 1$.

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Calibration Details: Hybrid Spender Model

- Without extending the model to include hand-to-mouth consumers, the quantitative exercise will not be able to fit well the data.
- μ is the proportion of HtM. $(1 - \mu)$ of consumers will behave exactly as before.
- Lump-sum transfers by government will guarantee same consumption in steady-state for all.
- Model unchanged with the exception of the demand block:

$$c_t = (1 - \beta\omega) \cdot d_t + [\mu + (1 - \mu)(1 - \beta\omega)] \cdot \left((y_t - t_t) + (1 - \mu)\mathbb{E}_t \left[\sum_{k=1}^{\infty} (\beta\omega)^k (y_{t+k} - t_{t+k}) \right] \right)$$

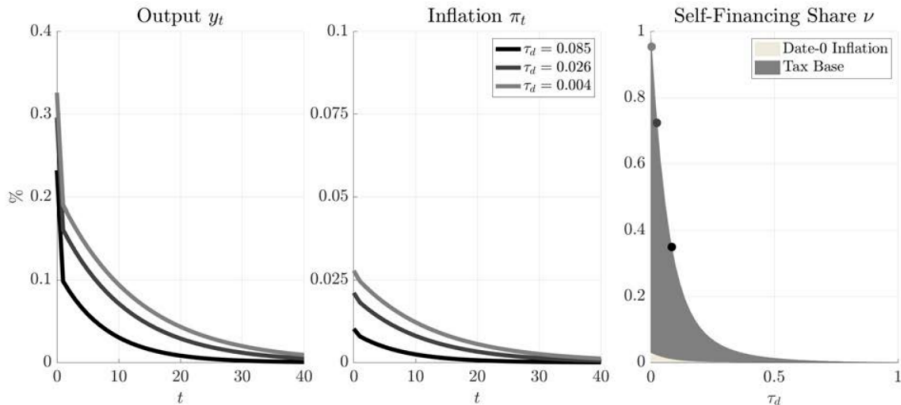
- Thus we allow MPC of income be larger than MPC of wealth.

Calibration Details: Referred Literature

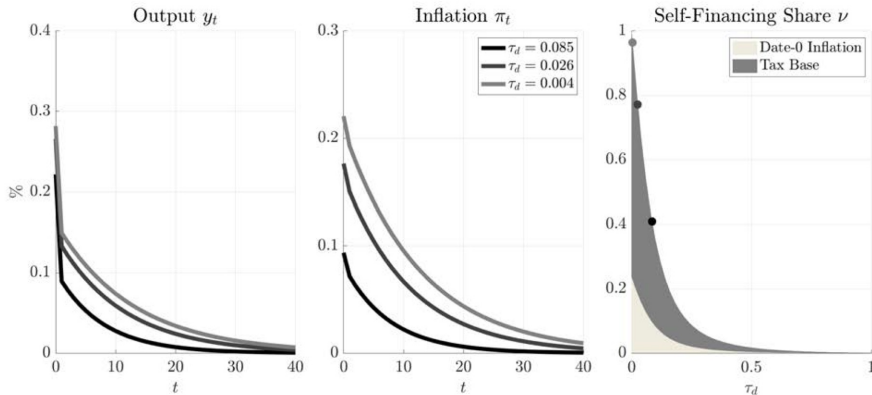
- **For ω and μ :** Based on Fagereng, Holm and Natvik (2021), empirical evidence on the household-level consumption response to income gains.
- **For τ_d :** The three values come from: Galí, López-Salido and Vallés (2007), Bianchi and Melosi (2017) and Auclert and Rognlie (2018).
- **For τ_y :** In line with DeLong and Summers (2012).
- **For κ :** Consistent with related literature like Galí and Gertler (1999), Christiano, Eichenbaum and Evans (2005), Hazell et al. (2022) and Barnichon and Mesters (2020).

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Previous Calibration with $\kappa = 0.01$



New Calibration with $\kappa = 0.1$ (steeper NKPC)



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Extensions of the Model

- 1 **More general demand structures:** Results will extend to more general demand structures as long as consumption preserves discounting and sufficient front-loading (i.e. HANK). [» Details](#)
- 2 **Investment:** scale effects but shape of dynamics are the same. [» Details](#)
- 3 **Distortionary tax hikes:** nothing essential changes. [» Details](#)
- 4 **Government spending:** similar dynamics but more quick effect. [» Details](#)
- 5 **More general monetary policy:** If monetary policy accomodates fiscal space the self-financing results plays-out even faster. If the monetary policy is hawkish enough, complete self-financing will be impossible (no bounded equilibria). [» Details](#)
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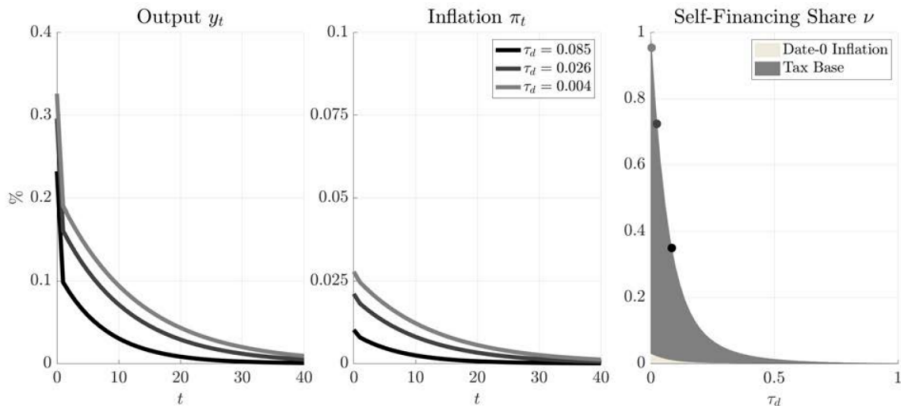
Extension: More General Demand Structure

- Demand block will be generalized:

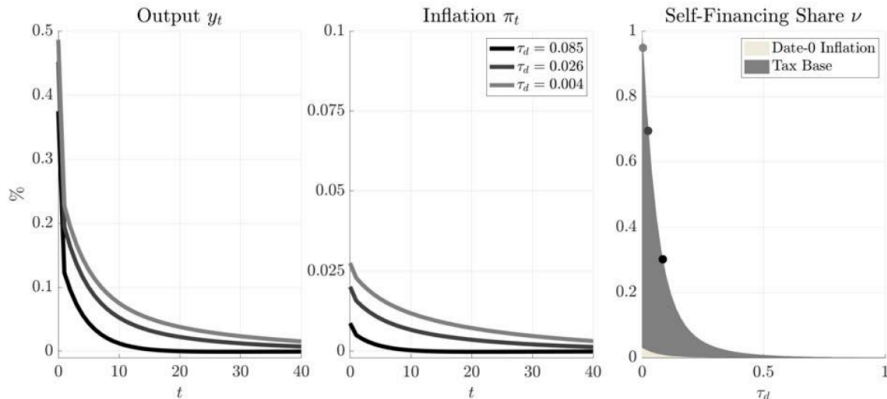
$$c_t = M_d \cdot d_t + M_y \cdot \left((y_t - t_t) + \delta \cdot \mathbb{E}_t \left[\sum_{k=1}^{\infty} (\beta\omega)^k (y_{t+k} - t_{t+k}) \right] \right)$$

- Here we are allowing different MPC for current income and wealth (generally we believe $M_d \leq M_y$). And also we disentangled geometric discounting from the contemporaneous MPC. Finally we allow for constant discounting rate δ .
- Theorem 2 which we will not develop establishes that if there is discounting and sufficient front-loading there will exist a unique bounded equilibrium where complete self-financing will be the limit fiscal adjustments infinitely delayed.
- This can nest model with liquidity constraints, hand-to-mouth agents, HANK or models with informational/behavioral frictions.

Our Simple Model Results



Full HANK Model



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Extension: Adding Investment

- Now firms will employ both L and K , with K owned and accumulated by the firm.
- Aggregate demand equation still holds but now in terms of c_t rather than y_t .
- Since investment is not taxed law of motion for real public debt still continues to hold now in terms of c_t rather than y_t .
- Proposition 1 still holds reinterpreting y_t as c_t .
- The change is now how the dynamics of investment and capital influence the real marginal costs and thereby date-0 inflation, but this only has a scale effect.
- The self-financing result still applies.

Extension: Distortionary Tax-Hikes

- Replace our fiscal rule with the following alternative:

$$T_{i,t} = \tau_{y,t} Y_{i,t} - \mathcal{E}_t$$

- Where our time-varying distortionary tax rate $\tau_{y,t}$ is:

$$\tau_{y,t} = \tau_y + \tau_d \frac{D_t - D^S S}{Y^S S}$$

- But after a log-lin this extended model will map into same aggregate demand and law of motion for real debt as before.
- The only difference is that the NKPC will have a new cost-push shock:

$$\pi_t = \kappa(y_t + \zeta d_t) + \beta \mathbb{E}_t[\pi_{t+1}]$$

- Just scale effects, our results are robust to the specification of the NKPC.

Extension: Model with Public Expenditure

- Now we have G_t , a stochastic, mean-zero spending shock, and shut down the lump-sum transfer from before. The new equations for the government will be:

$$d_{t+1} = \frac{1}{\beta} \left[d_t - t_t + g_t + \beta \frac{D^{SS}}{Y^{SS}} (i_t - \pi_{t+1}) \right]$$

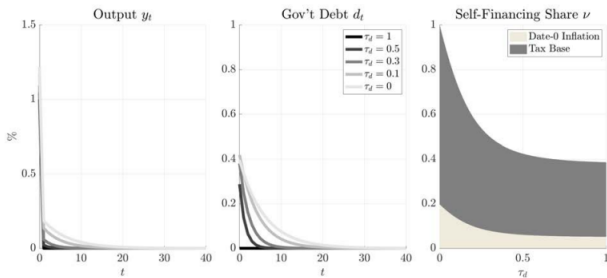
$$t_t = \tau_d \cdot (d_t + (1 - \tau_y)g_t) + \tau_y y_t$$

- And the the aggregate output market clearing condition:

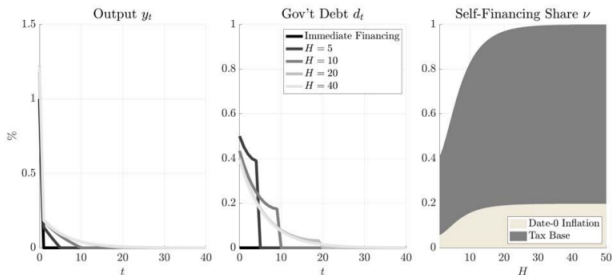
$$y_t = c_t + g_t$$

- Two effects, maps immediately one-to-one into demand, and g_t will add a cost-push shock to the NKPC. The second effect is immaterial. But first effect implies now even immediately tax financed government purchases have real effects.

BASELINE FISCAL POLICY



ALTERNATIVE FISCAL POLICY

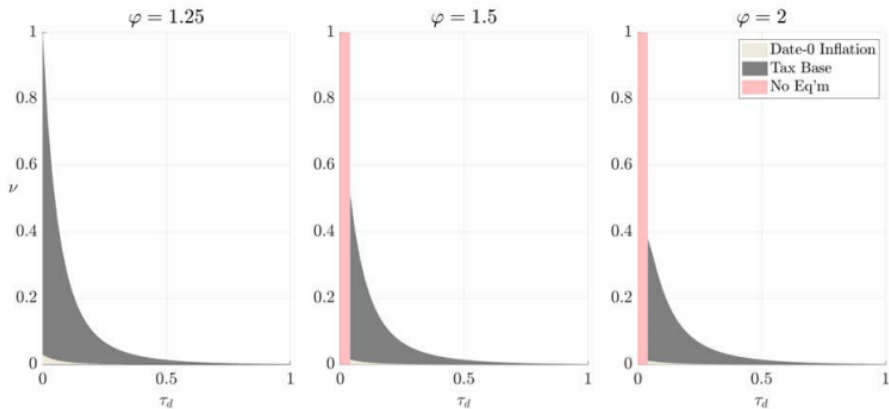


Extension: More General Monetary Policy

- We are still on a framework where:

$$r_t = \phi y_t$$

- $\phi > 0$ implies a hawkish central banker while $\phi < 0$ an accommodative one.
- Restrict $\phi \in [-\frac{1}{\sigma}, \frac{\tau_y}{\beta \frac{D}{Y} SS}]$. The lower bound is there because then deficit boom will be so large that there is immediate self-financing. The upper bound $\hat{\phi}$ the tax base channel will fully counteracted.
- **Theorem 3:** $\omega < 1$ and $\tau_y > 0$. There exist $\bar{\phi}$ within those bounds such that: if $\phi < \bar{\phi}$ complete self-financing will be possible while if $\phi > \bar{\phi}$ complete self-financing will not be possible (ν will be bounded by a number lesser than 1).



Relationship with pre-existing literature

- $r - g$: we share the similarity that deficits can be financed without future tax hikes. But we do a different channel. In our model $r > g$.
- **FTPL**: Some similar flavor but different mechanism. We shift the discussion from the debt erosion channel to the tax base channel. We avoid the equilibrium selection discussion. Maybe the FTPL idea is more robust than what was previously believed?
- **Fiscal Policy within NK models**: already well-understood that fiscal deficits can generate demand-led booms, thus also contributing with their own financing. Our paper characterizes the determinants of self-financing and it stresses the possibility of complete self-financing.
- **DeLong and Summers (2012)**: very similar exercise but without a microfounded model.